ASSIGNMENT- X

TUTORIAL XV

Q1. A city has four roads in each direction. Observed frequencies of the usage of the four different roads by 1000 automobiles on a particular day are recorded:

Road	1	2	3	4
Frequency	294	276	238	192

Using chi-square test, investigate whether drivers of automobiles preferred any particular road to others for travelling at the significance level $\alpha = 0.05$. (Given $P(\chi^2_3 > 7.81) = 0.05$.)

Q2. A survey of $n = 200$ families each with four children revealed the following frequency distribution of the number of girls per family.

No. of girls	0	1	2	3	4
No. of families	5	32	65	75	23

Test whether the data follows a Binomial distribution. (Note $P(\chi^2_3 > 7.81) = 0.05$).

Q3. Consider the confidence interval

$$\left[\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

for the parameter θ of the normal distribution with mean θ and known variance σ^2 with CI $100(1-\alpha)\%$. If we want a CI of prescribed length d, find n.

PTO

Q4. Determine the regression line which best fits the following points:

(x, y) : $(1,3)$, $(2,3)$, $(4,7)$, $(5,6)$, $(8,12)$

Q5. How do you measure the variation in the response or dependent y_i $1 \le i \le n$ by the regressor or independent x_i, $1 \le i \le n$?